

Reinforcement Learning

Monte Carlo Methods Sutton/Barto* Chapter 5

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With figures from Sutton/Barto*

*Sutton and Barto, Reinforcement Learning: An Introduction,
2nd edition, MIT Press, Cambridge, MA, 2018



Online Material



Topics of this Course

- Introduction to reinforcement learning
- Markov decision processes
- **Part I: Tabular Methods**
 - Dynamic programming
 - **Monte Carlo methods**
 - Temporal-difference learning
 - Multi-step bootstrapping
 - Planning and learning with tabular methods
- Part II: Approximate Solution Methods
 - Prediction and Control using Approximation
 - Eligibility Traces
 - Policy Gradient Methods
- Part III: Modern RL Methods
 - Deep Reinforcement Learning
 - Current Applications

Summary of Notation

General

X	capital letters: random variables
x, p	lower-case letters: realizations of random variables or scalar functions
$\mathbf{w}$	Bold lower-case letters: real-valued vectors (even if random variables)
$\mathbf{W}$	bold capitals: matrices
α	Greek letters: parameters (vectors if in bold)
$\Pr\{X = x\}$	probability that a random variable X takes on the value x
$X \sim p$	random variable X selected from distribution $p(x) = \Pr\{X = x\}$
$\mathbb{E}[X]$	expectation of a random variable X , i.e., $\mathbb{E}[X] = \sum_x p(x)x$
$\operatorname{argmax}_a f(a)$	a value of action a at which $f(a)$ takes its maximal value

Value Function

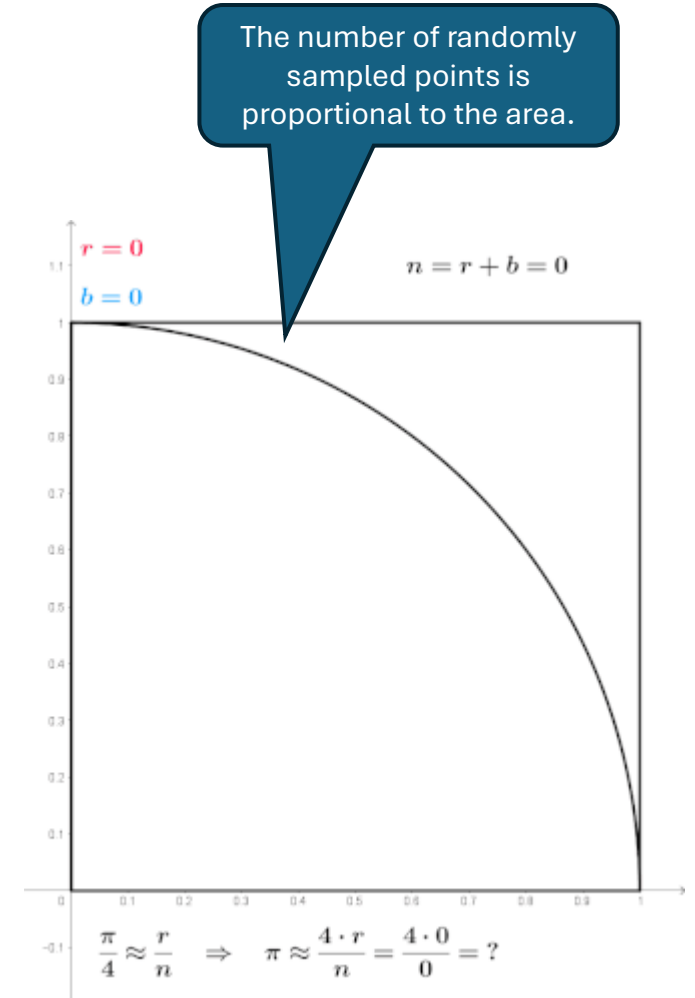
G_t	return (cumulative reward) following time t
$G_{t:h}$	return from t to h (discounted and corrected)
$v_\pi(s)$	value of state s under policy π (expected return)
$v_*(s)$	value of state s under the optimal policy
$q_\pi(s, a)$	value of taking action a in state s under policy π
$q_*(s, a)$	value of taking action a in state s under the optimal policy
V, V_t	array estimates of state-value function v_π or v_*
Q, Q_t	array estimates of action-value function q_π or q_*

MDP

s, s'	states
a	an action
r	a reward
$\mathcal{S}$	set of all (nonterminal) states, $\mathcal{S}^+$ are all states
$\mathcal{A}(s)$	set of all actions available in state s
γ	discount-rate parameter
t	discrete time step
T	final time step of an episode (a.k.a. horizon)
A_t	random variable for the action at time t
S_t	random variable for the state at time t
R_t	random variable for the reward at time t
$p(s', r s, a)$	probability of transition to state s' and receiving reward r , from state s taking action a .
$p(s' s, a)$	probability of transition to state s' from state s taking action a .
$r(s, a)$	expected immediate reward from state s after action a .
$r(s, a, s')$	expected immediate reward from state s to s' with action a .
$\pi(a s)$	probability of taking action a in state s under stochastic policy π
$\pi(s)$	action taken in state s under deterministic policy π

Introduction

- **Model-free method:**
 - We do not assume knowledge of the environment model (the MDP).
 - We use “experience” = **sample sequences** of states, actions, and rewards from actual or simulated interactions with the environment.
- **Remember from AI:** model-free means the agent faces an unknown, fully observable, possibly stochastic environment.
- **Advantages:**
 - We often have **sample access** to systems.
 - It is easier to simulate transitions than to specify the complete MDP.
- **Monte Carlo (MC) methods** are a family of algorithms that rely on repeated random sampling to obtain numerical results (see example to the right).
- **Idea:** Use MC to sample sequences from the environment and use them to estimate the value function.



Monte Carlo method applied to approximating the value of π

Source: Wikipedia

Monte Carlo Prediction

Estimate state values by sampling with a given policy.

MC Value Function Approximation

- Use the sample-average method to approximate the expectation in:

$$v_{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid S_t = s]$$

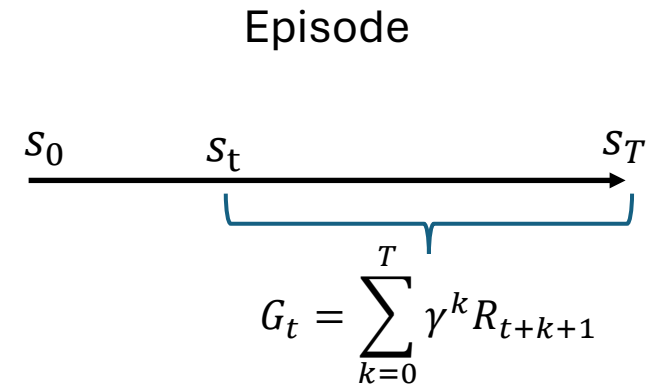
- **Convergence:** The average return of many sampled sequences (called episodes or trajectories) starting at s and following π converges to the true expectation as the number of samples increases.

$$v_{\pi}(s) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n G_{t|S_t=s,\pi}$$

- **Properties:**
 - **High Variance:** Sampling a single episode return has high variance since it contains the randomness over the whole sampled episode. Averaging over many sampled returns reduces variance.
 - **Unbiased:** The sample-average method provides an unbiased estimate (is on average correct) and thus converges for large n .
 - **Not a true online method:** This method needs to sample whole episodes before any calculation can be done. It is not a true step-by-step online method!

MC Prediction: Estimate v_π for a given π

- **Idea:** We can update all state values along the episode.
- **First-visit:** Using the return only for the first visit of a state in the episode ensures that the sampled returns are i.i.d. distributed (not dependent on each other) and we get an **unbiased** estimate.



← Calculate the return following s_t for all states in the episode

First-visit MC prediction, for estimating $V \approx v_\pi$

Input: a policy π to be evaluated

Initialize:

$V(s) \in \mathbb{R}$, arbitrarily, for all $s \in \mathcal{S}$

$Returns(s) \leftarrow$ an empty list, for all $s \in \mathcal{S}$

Loop forever (for each episode):

Generate an episode following π : $S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless S_t appears in $S_0, S_1, \dots, S_{t-1}$:

Append G to $Returns(S_t)$

$V(S_t) \leftarrow \text{average}(Returns(S_t))$

We observe a sequence of rewards

Calculate remaining returns in reverse order!

Uses only the first visit.

This sample average approximates the expected return.
Note: This can be implemented with incremental updating of the averages!

Monte Carlo Control

Find a good policy while using it to sample the data.

MC Control as Generalized Policy Iteration (GPI)

- Approx. π_* using ideas from GPI with a Q-function estimated using MC prediction.

- MC policy iteration:

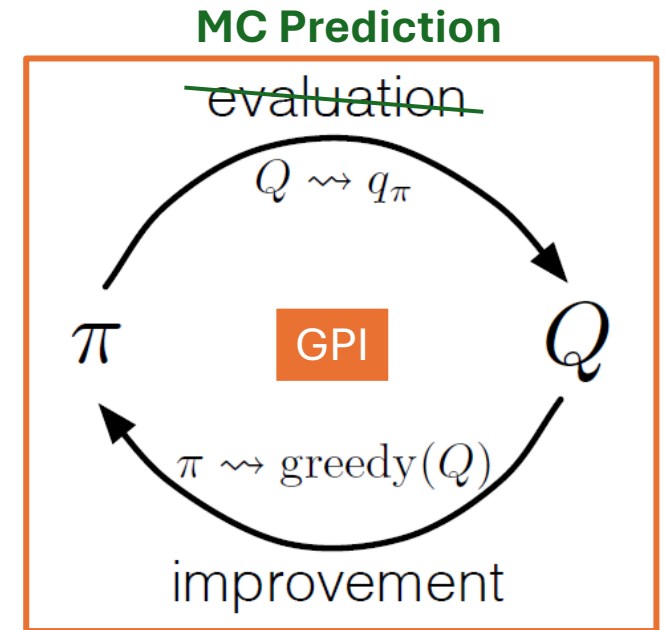
$$\pi_0 \xrightarrow{E} q_{\pi_0} \xrightarrow{I} \pi_1 \xrightarrow{E} q_{\pi_1} \xrightarrow{I} \pi_2 \xrightarrow{E} q_{\pi_2} \xrightarrow{I} \pi_3 \xrightarrow{E} q_{\pi_3} \xrightarrow{I} \dots \pi_* \xrightarrow{E} q_*$$

- Process:

1. Start with an arbitrary policy.
2. **Evaluation:** Update MC estimate by sampling some episodes using the current π .
3. **Improvement:** greedy(Q) means

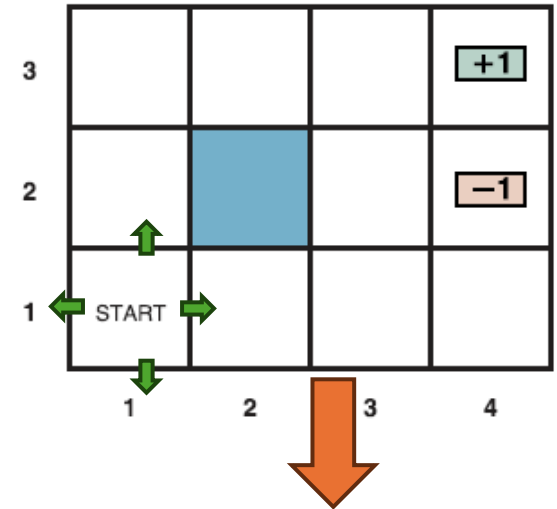
$$\pi(s) = \operatorname{argmax}_a q(s, a) \quad \forall s \in \mathcal{S}$$

4. Repeat evaluation and improvement till the policy does not change (= optimal policy is reached).



MC Prediction to Estimate Q-Values

- MC prediction for $q_{\pi}(s, a) = \mathbb{E}_{\pi}[G_t \mid S_t = s, A_t = a]$.
- Same as for $v_{\pi}(s)$, just record and average returns for action-state pairs (a, s) when following π .
- **Convergence:** Unbiased samples lead to convergence to q_{π} after ∞ episodes.
- **Exploration issue:** Many (a, s) -pairs that are important for π^* may never be visited when following π . To get Q-Value estimates, we need to keep exploring!
- Options to guarantee that all (a, s) -pairs are sampled :
 - Exploring starts:** choose the starting (a, s) for each episode randomly.
 - Only consider **soft policies** like ϵ -greedy. Remember: Soft policies have non-zero probabilities for all actions so they will explore.



Q-Table

s	a	$q(s, a)$
(1,1)	up	Sample average
(1,1)	right	Sample average
(1,1)	down	Sample average
(1,1)	left	Sample average
...	...	...

Remember: Policy Improvement Theorem

If we update the MC $q(s, \cdot)$ estimate by adding more unbiased sample returns following π_k , then we can construct a new greedy policy π_{k+1} that is better (or equal) compared to π_k . Reason:

$$\begin{aligned} v_{\pi_{k+1}}(s) &= q_{\pi_k}(s, \pi_{k+1}(s)) = q_{\pi_k}(s, \operatorname{argmax}_a q_{\pi_k}(s, a)) \\ &= \max_a q_{\pi_k}(s, a) \\ &\geq q_{\pi_k}(s, \pi_k(s)) \\ &\geq v_{\pi_k}(s) \end{aligned}$$

π_{k+1} chooses greedy action for s

Choosing the best action will never lead to a lower value!

→ converges to π_*

Assumptions

- All state/action pairs are tried (e.g., start episodes with exploring states)
- Perfect policy evaluation with ∞ many episodes before each policy update. **Perfect policy evaluation is impractical!**

The effect of imperfect policy evaluation

- Individual q estimates will be wrong but unbiased return estimates will produce in expectation (on average) the correct estimate.
- Policy improvement sometimes may pick the wrong action, but **on average favors actions with true higher values.**

Alg. MC Control with Exploring Starts

First-visit MC prediction, for estimating $V \approx v_\pi$

Input: a policy π to be evaluated

Initialize:

$V(s) \in \mathbb{R}$, arbitrarily, for all $s \in \mathcal{S}$

$Returns(s) \leftarrow$ an empty list, for all $s \in \mathcal{S}$

Loop forever (for each episode):

Generate an episode following π : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode:

$G \leftarrow \gamma G + R_{t+1}$

Unless S_t appears in $S_0, S_1, \dots, S_{t-1}$:

Append G to $Returns(S_t)$

$V(S_t) \leftarrow \text{average}(Returns(S_t))$

So all state-action pairs are tired.

Monte Carlo ES (Exploring Starts), for estimating $\pi \approx \pi_*$

Initialize:

$\pi(s) \in \mathcal{A}(s)$ (arbitrarily), for all $s \in \mathcal{S}$

$Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

$Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

Loop forever (for each episode):

Choose $S_0 \in \mathcal{S}, A_0 \in \mathcal{A}(S_0)$ randomly such that all pairs have probability > 0

Generate an episode from S_0, A_0 , following π : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1, \dots, S_{t-1}, A_{t-1}$:

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$\pi(S_t) \leftarrow \arg\max_a Q(S_t, a)$

Learn policy

Q -Values instead of V .

Exploring starts

First-visit

Greedy policy update

Code...

MC Control without Exploring Starts

- **Issue:** The environment needs to let us start episodes at different (s, a) at will. This is fine for simulations, but hard in the real world.
- **Need:** For convergence, we need a guarantee that all (reachable) state-action combinations are tried.
- Options:
 - a) **On-policy control:** learn a **soft policy** instead of the optimal policy.
 - b) **Off-policy control:** use a soft **behavior policy** for sampling that tries all state-action pairs, but learn the optimal, deterministic **target policy**.

On-Policy Monte Carlo Control

Find a good policy while using it to sample.

Exploration with On-Policy Methods

On-policy methods learn the policy they follow.

Dilemma:

- We need behavior (follow a policy) that explores. Such a policy is called a soft policy.
- We know that MDPs have optimal deterministic policies that do not explore.

Optimality issue: On-policy methods can only learn soft policies. We can convert a soft policy into a greedy deterministic policy, but it may not be optimal!

Types of Soft Policies for Exploration

Definition : A policy is soft if
 $\pi(a|s) > 0 \quad \forall s \in \mathcal{S}, a \in \mathcal{A}$

Soft policies are stochastic and used as behavior policies that guarantee exploration.

Important Soft Policy Types:

- **ϵ -soft policy:** agent takes with probability ϵ a random action.

$$\pi(a|s) \geq \frac{\epsilon}{|\mathcal{A}(s)|}$$

- **ϵ -greedy policy:** follows a greedy policy but chooses a random action with probability ϵ

$$\pi(a|s) = \begin{cases} 1 - \epsilon + \frac{\epsilon}{|\mathcal{A}(s)|} & \text{if } a = \text{greedy action} \\ \frac{\epsilon}{|\mathcal{A}(s)|} & \text{if } a \neq \text{greedy action} \end{cases}$$

Alg. On-Policy MC Control with ϵ -soft Policies

Monte Carlo ES (Exploring Starts), for estimating $\pi \approx \pi_*$

Initialize:

$\pi(s) \in \mathcal{A}(s)$ (arbitrarily), for all $s \in \mathcal{S}$

$Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

$Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Loop forever (for each episode):

Choose $S_0 \in \mathcal{S}$, $A_0 \in \mathcal{A}(S_0)$ randomly such

Generate an episode from S_0, A_0 , follow

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in S_0, A_0 ,

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$\pi(S_t) \leftarrow \arg\max_a Q(S_t, a)$

On-policy first-visit MC control (for ϵ -soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\epsilon > 0$

Initialize:

$\pi \leftarrow$ an arbitrary ϵ -soft policy

$Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

$Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Repeat forever (for each episode):

Generate an episode following π : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1, \dots, S_{t-1}, A_{t-1}$:

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$A^* \leftarrow \arg\max_a Q(S_t, a)$

(with ties broken arbitrarily)

For all $a \in \mathcal{A}(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \epsilon + \epsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \epsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

Create ϵ -greedy policy

Important: On-policy learns the best ϵ -greedy policy, but not the optimal deterministic policy!

GLIE: Greedy in the Limit with Infinite Exploration

Issues:

1. On-policy MC control algorithms change the policy and **average over returns from different policies**. This could be a problem for convergence!
2. On-policy methods learn a **soft policy**.

GLIE Condition

Many on-policy control methods converge to an optimal deterministic policy if:

1. **Infinite explorations:** Every state-action pair is visited infinitely often.

$$\lim_{t \rightarrow \infty} N_t(s, a) = \infty$$

2. **Greedy in the limit:** The policy becomes greedy with respect to Q over time.

$$\lim_{t \rightarrow \infty} \pi(\operatorname{argmax}_a (Q_t(s, a), s)) = 1$$

Implementation:

- Infinite exploration: Use many episodes and an ϵ -greedy policy for exploration. This will “wash out” early “bad” observations.
- Greedy in the limit: Use an ϵ -greedy policy where ϵ is reduced over time. E.g., $\epsilon_t = \frac{1}{t}$

Off-Policy Prediction

Estimate a value function for a policy different from the sampling policy.

On-Policy vs. Off-Policy Control

On-policy: On-policy methods learn the policy they follow. These need to be exploring (soft).

Off-policy: Use two policies

- **Target policy:** learn π , which will become an optimal deterministic policy.
 - **Behavior policy** b : guarantees exploration with a soft policy.
- Called “off-policy” because the generated data follows a different policy than the learned target policy.
 - Off-policy methods are typically:
 - More complicated
 - Converge slower
 - Often more powerful
 - Can be applied to data externally generated using a fixed, potentially unknown, behavior policy.

Note: On-policy learning is a special case of off-policy learning with $b = \pi$

Off-Policy Prediction

- **Goal:** Estimate v_π or q_π given episodes generated with $b \neq \pi$.
- **Assumptions:**
 - b is fixed.
 - **Coverage:** Every action taken under π will occasionally also be taken under b

$$\pi(a|s) > 0 \Rightarrow b(a|s) > 0$$

- That is, b needs to be stochastic (soft) for states where it differs from π (typically b is ϵ -greedy)
- **Issue:** Returns and sample averages are valid for b but not for π ! We need to correct them.

Importance Sampling

- **Importance sampling** is a method to estimate the expected values under a desired distribution, given samples from another distribution

- Expectation under desired distribution: $\mathbb{E}_\pi(G_t|S_t = s)$
- Observed expectation from sampled distribution: $\mathbb{E}_b(G_t|S_t = s)$

- **Idea:** correct returns observed under b to estimate returns for π

- Probability of a trajectory starting at S_t under π

$$\Pr\{A_t, S_{t+1}, A_{t+1}, \dots, A_{T-1}, S_T | S_t, A_{t:T-1} \sim \pi\} = \prod_{k=t}^{T-1} \pi(A_k | S_k) p(S_{k+1} | S_k, A_k)$$

Needs unknown transition probabilities!

- **Importance sampling ratio:** How much more likely are the remaining actions in the sequence chosen under π than under b :

$$\rho_{t:T-1} \stackrel{\text{def}}{=} \frac{\prod_{k=t}^{T-1} \pi(A_k | S_k) p(S_{k+1} | S_k, A_k)}{\prod_{k=t}^{T-1} b(A_k | S_k) p(S_{k+1} | S_k, A_k)} = \prod_{k=t}^{T-1} \frac{\pi(A_k | S_k)}{b(A_k | S_k)}$$

Luckily, the unknown transition probabilities cancel out!

- Now we get:

$$v_\pi(s) = \mathbb{E}_\pi(G_t | S_t = s) = \mathbb{E}_b[\rho_{t:T-1} G_t | S_t = s]$$

- **Note:** We only get $\rho_{t:T-1} > 0$ when all A_k chosen by b have a $\pi(A_k | S_k) > 0$!

This especially a problem if π is a deterministic policy very different from b !

Ordinary vs. Weighted Importance Sampling

- **Ordinary importance sampling** averages the corrected sample returns

$$V(s) \stackrel{\text{def}}{=} \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{|\mathcal{T}(s)|} \rightarrow \mathbb{E}_b[\rho_{t:T-1} G_t | S_t = s] = \mathbb{E}_\pi(G_t | S_t = t)$$

- $\mathcal{T}(s)$ are the time steps that s was (first) visited.
- Gives an unbiased estimate of $v_\pi(s)$ but the variance is extreme for low $|\mathcal{T}(s)|$.

Note: the $\frac{|\mathcal{T}(s)|}{|\mathcal{T}(s)|}$ is missing since it canceled out.

- **Weighted importance sampling**

$$V(s) \stackrel{\text{def}}{=} \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1}} \rightarrow \frac{\mathbb{E}_b[\rho_{t:T-1} G_t | S_t = s]}{\mathbb{E}_b[\rho_{t:T(t)-1} | S_t = s]} = \frac{\mathbb{E}_\pi(G_t | S_t = t)}{1}$$

- Set 0 if denominator is 0.
- Initially biased to $v_b(s)$. The ρ terms cancel for a single observation and G_t is sampled using b .
- Bias falls asymptotically with more visits
- Preferred method with much lower errors (variance) for smaller number of episodes.

Alg. Off-Policy Prediction

Weighted importance sampling + incremental average update

On-policy first-visit MC control (for ε -soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\varepsilon > 0$

Initialize:

$\pi \leftarrow$ an arbitrary ε -soft policy

$Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

$Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

Repeat forever (for each episode $i = 1, 2, \dots$):

Generate an episode following π :

$G \leftarrow 0$

Loop for each step of episode, $t = 0, 1, \dots, T-1$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $Returns(S_t, A_t)$:

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$A^* \leftarrow \arg\max_a Q(S_t, a)$

For all $a \in \mathcal{A}(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \varepsilon + \varepsilon / |\mathcal{A}(S_t)| & \text{if } a = A^* \\ \varepsilon / |\mathcal{A}(S_t)| & \text{otherwise} \end{cases}$$

Off-policy MC prediction (policy evaluation) for estimating $Q \approx q_\pi$

Input: an arbitrary target policy π

Initialize, for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$:

$Q(s, a) \in \mathbb{R}$ (arbitrarily)

$C(s, a) \leftarrow 0$

cumulative sum of weights for (s, a) for incremental update of the average return.

Loop forever (for each episode):

$b \leftarrow$ any policy with coverage of π

b can change every episode! Practical choice is $b = \varepsilon\text{-greedy}(\pi)$

Generate an episode following b : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

$W \leftarrow 1$

Importance sampling ratio ρ

Loop for each step of episode, $t = T-1, T-2, \dots, 0$, while $W \neq 0$:

$G \leftarrow \gamma G + R_{t+1}$

$C(S_t, A_t) \leftarrow C(S_t, A_t) + W$

$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$

Can only learn from the end of the episode where the action chosen from b can also be chosen by π !

$W \leftarrow W \frac{\pi(A_t|S_t)}{b(A_t|S_t)}$

Incremental update + Correct the update

Off-Policy Control

Find a good policy while following a different behavior policy.

Off-Policy MC Control

- **Remember:** on-policy control tries to estimate a policy while using it for sampling. This has the following implications:
 - The **behavior changes** with the learned policy.
 - The policy that is learned needs to explore and therefore **may not be able to learn the optimal deterministic policy**.
- Off-policy control uses two policies:
 - **Behavior policy b :** Needs to have coverage. That is, it eventually takes every action taken under π . This typically means that we will use a **soft policy**.
 - **Target policy π :** Typically, a **deterministic greedy policy** which can become optimal.

Alg. Off-Policy MC Control

Off-policy MC prediction (policy evaluation) for estimating $Q \approx q_\pi$

Input: an arbitrary target policy π

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:

$Q(s, a) \in \mathbb{R}$ (arbitrarily)

$C(s, a) \leftarrow 0$

Loop forever (for each episode)

$b \leftarrow$ any policy with coverage

Generate an episode following b .

$G \leftarrow 0$

$W \leftarrow 1$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

$C(S_t, A_t) \leftarrow C(S_t, A_t) + W$

$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$

$W \leftarrow W \frac{\pi(A_t|S_t)}{b(A_t|S_t)}$

Can only use the end of the episode that uses greedy actions for π

Issue: Not very sample efficient!

Off-policy MC control, for estimating $\pi \approx \pi_*$

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:

$Q(s, a) \in \mathbb{R}$ (arbitrarily)

$C(s, a) \leftarrow 0$

$\pi(s) \leftarrow \operatorname{argmax}_a Q(s, a)$ (with ties broken consistently)

Add the greedy target policy

Loop forever (for each episode):

$b \leftarrow$ any soft policy

Generate an episode using b : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

$W \leftarrow 1$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

$C(S_t, A_t) \leftarrow C(S_t, A_t) + W$

$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$

$\pi(S_t) \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken consistently)

If $A_t \neq \pi(S_t)$ then exit inner Loop (proceed to next episode)

$W \leftarrow W \frac{1}{b(A_t|S_t)}$

We use π to select a . So the probability is 1.

What you Should Know



MC works with sampled episodes:

- **Policy Evaluation:** Averages many sampled returns.
- **Control:** Interleaves pol. eval. and pol. improvement on an episode-by-episode basis.
- **On vs. off-policy control.**

Advantages of MC methods vs. DP:

- **Model-free:** No need for an environment model. Sampling from a simulation is often easy.
- Can evaluate a small **state set** of interest. DP always has to sweep over all states.
- **Unbiased estimates:** Estimates are based on returns from complete sampled sequences. We will learn about other methods that introduce bias by bootstrapping (estimating a value from other estimates).
- Using complete sequences may be less harmed by violations of the Markov property/issues with observability.

Issues:

- **Needs to maintain sufficient exploration.** Options are:
 1. Exploring starts can learn the optimal policy.
 2. On-policy control with an ϵ -soft policy and GLIE.
 3. Off-policy control with a covering behavior policy.
Note: only a part of the data can be used.
- MC methods are **not sample efficient**. I.e., sampled episodes are used only once and then discarded.
- **Not a truly online method:** Updates are not done after each action, but are delayed until the end of each episode.